

Seth Baird

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EDUCATION

- **Purdue University** West Lafayette, Indiana
Anticipated Graduation May 2027 *GPA 3.09*
Major in Computer Engineering Technology Minor in Information Technology
- **Memorial High School** Frisco, Texas
Graduated May 2023 *GPA 3.81*
Vocational Education: CS3: Data Structures and Algorithms, Mobile Application Development

SKILLS

C | STM32 | Firmware Engineering | PLC | FPGA | CAD | PCB Design | Circuit Analysis | Python | Java

PROJECTS

- Designed and built a miniature guitar amplifier with a CAD-drawn case and a purpose-built PCB in six weeks. My team and I simulated the circuit in Multisim and created frequency response curves in lab.
- Modernized an analog superheterodyne FM radio with automated channel scan, presets, and a display. Using a local maxima locator algorithm, it searches the continuous stream of RSSI data for strong signals.
- Built industrial control systems hands-on in lab and programmed PLCs using the IEC 61131-3 standards.
- Created and simulated concurrent logic circuits and HDL code for Altera FPGAs in Quartus.
- Developed an android app that integrates with a car's engine computer to provide insights to the driver.

RELEVANT COURSEWORK

Prototyping & Development | Industrial Controls | Data Structures & Algorithms | Advanced Embedded Systems

EXPERIENCE

- **Purdue Electric Racing: Software Team**
 - Reworked the CAN communication backend to use two busses simultaneously for new high-bitrate motors.
 - Developed the PWM HAL (hardware abstraction layer) to control onboard variable-speed devices like cooling fans. This allowed the speed to be modulated in two lines instead of 50, allowing for easier addition of new hardware.
 - Leveraged specialized testing equipment such as CANable and logic analyzers to verify successful implementation of mission critical systems.
- **SMS InfoComm (Wistron Corporation)** Dallas, TX
NVIDIA Server Repair Technician *05/2024 - 08/2024*
 - **System Repair and Troubleshooting** Diagnosed, refurbished, and built NVIDIA HGX 8-GPU units.
 - **Hardware Validation and Analysis** Used proprietary diagnostic tools to identify errors and discrepancies within motherboard subsystems such as power delivery, signal integrity, and FPGAs before executing a repair.
 - **Team Workflow Optimization** Developed a Python script to automate thousands of lines of by-hand data entry into the company's antiquated database, saving my team 80 hours of labor per week.
- **Jawa.gg** Remote
Verified Seller *10/2021 - 08/2023*
 - **Top 10 Seller in Overall Performance** Recognized by CEO Amanda Stefan for driving Jawa's growth.
 - **Computer Sales and Commissions** Built and sold dozens of tailored desktop, laptop, and Mac computers to end users at different price points and configurations.
 - **Part Refurbishment and Sales** Repaired, tested, and refurbished parts and sold them to PC builders and end users through Jawa's marketplace.